

Anela K. Akiona

San Diego, CA / (808) 386-6532 / aakiona@ucsd.edu

A passionate marine ecologist with expertise in research study design and execution, fish data collection and analysis, scientific diving, statistical analysis and coding, coral reef ecology, and science communication, with a strong commitment to fostering inclusive research.

Research Interests

My research aims to describe changes in coral and fish communities over time, investigate methods to improve ecosystem resilience to climate change, support and empower local communities, and facilitate access to technologies and research.

Education

PhD: Scripps Institution of Oceanography, UC San Diego **Sep 2018 – Jan 2026**

- PhD in Marine Biology in Dr. Stuart Sandin's lab, with a focus on coral and fish ecology.
- GPA: 3.976/4.0
- Thesis: *Community ecology and resilience of coral reefs*
 - Ch. 1: Using a decision-support model to map the efficacy of coral interventions in the Maldives
 - Ch. 2: Impacts of herbivory on coral competition and resilience
 - Ch. 3: Length-weight relationships for 18 coral reef fish species from the central Pacific
 - Ch. 4: Length-weight relationships for 15 coral reef fish species from the Northern Line Islands
 - Ch. 5: A database of life history parameters for Pacific coral reef fish

MS: University of Hawai'i at Mānoa **Aug 2016 – Aug 2018**

- MS in Marine Biology in Dr. Erik Franklin's lab, with a focus on fish population dynamics.
- GPA: 3.88/4.0
- Thesis: *Predatory fish population dynamics and diet in a traditional Hawaiian fishpond*
 - Ch. 1: Fishing for Science: Assessing predatory fish populations in He'eia Fishpond
 - Ch. 2: Diets of the predatory fish of He'eia Fishpond: Insights from stomach content and stable isotope analyses

BA: University of San Diego **Sep 2010 – May 2014**

- Bachelor of Arts in Marine Science (Biology pathway), minor in Mathematics
- GPA: 3.61/4.0 (graduated *Cum Laude*)

School for Field Studies, South Caicos, Turks and Caicos Islands **Feb 2013 – May 2013**

- Study abroad program focused on tropical marine ecology, including SCUBA-based research methods.
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Professional Experience

Scripps Institution of Oceanography, Graduate Student Researcher (La Jolla, CA) **Sept 2018 – Jan 2026**

- Lead an independent research team within a multi-organizational project, including groups from NOAA, Florida Institute of Technology, and UH Mānoa. Developed original research questions within the group's overarching goal of developing place-based decision support models for coral restoration.
- Supervised the fish research group within the lab, including organizing meetings, managing projects, and spearheading writing and presentations.
- Assisted in lab projects, including large-area imagery and fish survey data collection and analysis, training, and coding.

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- Facilitated lab meeting and present regularly, participated in lab outreach events to get the public and local school kids interested in science.
- Led and participated in student committees to increase the diversity of the Scripps student body.

University of Hawai‘i at Mānoa, Graduate Student Researcher (Honolulu, HI)

Aug 2016 – Aug 2018

- Developed a research project with a local NGO (Paepae o He‘e‘ia) to assess the population of predatory fish in a traditional Hawaiian fishpond working towards food production.
- Synthesized and presented research in scientific writing, conference presentations, and local outreach events.

Hawai‘i Institute of Marine Biology, Research Technician (Kāne‘ohe, HI)

Jan 2016 – Aug 2016

- Assisted in the design and execution of field and laboratory research, oversaw lab safety protocols, and managed supplies and equipment.
- Facilitated new research and collaboration with He‘e‘ia Fishpond, which led to MS research.

School for Field Studies, Waterfront Assistant (Turks and Caicos Islands)

Aug 2014 – Aug 2015

- Participated in and supervised field research on South Caicos and surrounding islands including benthic assessments via SCUBA (fish survey, coral bleaching, benthic characterization, photo analysis using CPCe), beach profiling for turtle nesting, and water quality monitoring.
- Assisted faculty in overseeing students’ directed research, including finfish stock assessment, shark habitat characterization, eDNA collection, and BRUV deployment and recovery.
- Led species identification lectures both in the classroom and in the water.
- Managed all water-based research and recreational activities, including leading SCUBA and skin diving.
- Assisted DSO in instruction of PADI Open Water and Advanced Open Water courses.
- Responsible for operation and maintenance of all waterfront facilities, including SCUBA equipment and the Center’s five dive vessels.

Student Support Services, Academic Tutor (San Diego, CA)

Oct 2011 – Dec 2012

- Tutored undergraduate peers in various Biology, Chemistry, and Calculus classes.
- Worked with students to set attainable academic goals and successfully earn high achievement scores on homework and exams.

Research Experience (leader)

Scripps Institution of Oceanography, Graduate Student Researcher (La Jolla, CA)

Sept 2018 – Present

- Developed a spatially explicit model to investigate the impact of coral restoration in the Maldives. Worked with the Maldives Marine Research Institute (MMRI) to identify feasible future restoration methods. Presented findings of this work to collaborators at MMRI and NOAA, as well as at national and international conferences.
- Spearhead fish research group in compilation of a publicly accessible, quality-controlled length-weight parameter database. Lead project management and manuscript writing.

Hawai‘i Institute of Marine Biology, Researcher (Kāne‘ohe, HI)

Aug 2016 – Aug 2018

- Worked with managing organization of a traditional Hawaiian fishpond to determine areas of research that would be of use to them, and designed a research project around those goals.
- Organized teams of local fishers to conduct a mark-recapture experiment that sampled the entirety of the 88-acre pond across two days.

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- Attended community fishing days over the course of two years to measure predatory fish caught in the fishpond.
- Conducted a comprehensive dietary study on captured predatory fish, including visual gut content identification, stable isotope analysis, and genetic metabarcoding of prey items.
- Presented the findings of my work both at community events and scientific conferences.

School for Field Studies, Waterfront Assistant (Turks and Caicos Islands)

Aug 2014 – Aug 2015

- Led research project on invasive lionfish (*Pterois volitans*), including SCUBA surveys, morphometric measurements, dissections, and gut content analysis.
- Responsible for the coordination and execution of a beach erosion study across three islands.
- Authored an article for a local magazine about the Center's work and research on lionfish for dissemination to the broader Turks and Caicos community.

Hawai'i Division of Aquatic Resources, NSF REU Intern (Hilo, HI)

Jun 2013 – Aug 2013

- Generated length-weight relationships (LWR) for common Hawaiian estuarine fishes. Utilized a cast net to capture fish.
- Analyzed long-term fish measurements from estuary sampling to determine patterns of biomass and abundance.
- The LWR from my research is still used by DAR for biomass calculations.

Pacific Islands Climate Change Cooperative, Research Intern (Honolulu, HI)

Jun 2012 – Aug 2012

- Designed and implemented a vulnerability assessment of native Hawaiian plants and birds to adverse climate change effects.
- Conducted a literature review to determine which traits cause a species to be most at risk. Presented findings to PICCC's steering committee consisting of representatives from Hawai'i's major conservation organizations.

Research Experience (supporting)

100 Island Challenge, Researcher (Maui Nui, Palau, Samoa, the Mariana Islands, Taiwan)

May 2019 – Jun 2025

- Participated in fish belt transect surveys (Samoa, the Mariana Islands) and large-area imagery surveys (Maui Nui, Palau, Taiwan) as part of the 100 Island Challenge, an effort to document natural variation in coral reefs.
- Compiled and disseminated fish identification study materials and co-led review sessions.
- Taught local and governmental partners fish survey methods and large-area imagery procedures.

Curaçao Long-term Monitoring Program, Researcher (Curaçao)

Oct 2023 – Nov 2023

- Conducted fish belt transect surveys across the leeward coast of Curaçao as part of a long-term CARMABI study.

Waitt Institute, Researcher (Palau, Samoa, Fiji, Vanuatu)

Jan 2022 – Sep 2023

- Conducted ship-based fish belt transect surveys across five expeditions totaling 13 weeks at sea.
- Trained NGO and governmental partners on fish and invertebrate survey methods.

Island Ocean Connection Challenge, Researcher (Palau)

Feb 2023

- Co-led a workshop for local partners (One Reef and the Hatohobei State Rangers) on fish survey methods and identification that culminated in practice SCUBA surveys.
- Conducted fish surveys on two islands as part of a rat eradication study with Island Conservation.

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Publications

Refereed:

- Akiona, A.K.**, B.J. Zgliczynski, M.M. Agarwal, B.J. French, N. Hanna Holloway, K.A. Lubarsky, M.E. Shirley, C.J. Sullivan, S.A. Sandin. 2025. A database of life history parameters for Pacific coral reef fish. *Scientific Data*, 12(1425). <https://doi.org/10.1038/s41597-025-05788-x>
- Akiona, A.K.**, B.J. Zgliczynski, B.J. French, S.A. Sandin. 2025. Length-weight relationships for 15 coral reef fish species from the Northern Line Islands. *Pacific Science*, 78(4):399-406. <https://doi.org/10.2984/78.4.4>
- Cacapit, A.J., **A.K. Akiona**...S.A. Sandin. 2025. Initial coral assemblage drives benthic community response to different disturbance type events. *PLOS ONE*, 20(5): e0317515. <https://doi.org/10.1371/journal.pone.0317515>
- McCarthy, O.S., E.L.A. Kelly, **A.K. Akiona**, S.M. Clements, T. Martinez, N.E. Pedersen, C. Peralto, S.L. Romero, M.H. Smelser, K.W. Stone, R.T. Sparks, J.E. Smith. 2025. Using community composition and successional theory to guide site-specific coral reef management. *Global Change Biology*, 31:e70050. <https://doi.org/10.1111/gcb.70050>
- Akiona, A.K.**, B.N. Popp, R.J. Toonen, M.C. Siple, K. Kotubetey, H. Kawelo, E.C. Franklin. 2022. Predatory fish diet shifts towards an invasive mullet in a traditional Hawaiian aquaculture system. *Aquaculture Fish and Fisheries*, 2:307-320. <https://doi.org/10.1002/aff2.68>
- Akiona, A.K.**, B.J. Zgliczynski, S.A. Sandin. 2021. Length-weight relationships for 18 coral reef fish species from the central Pacific. *Journal of Applied Ichthyology*, 00, 1-5. <https://doi.org/10.1111/jai.14249>
- Peyton, K.A., T.S. Sakihara, L.K. Nishiura, T.T. Shindo, T.E. Shimoda, S. Hau, **A. Akiona**, K. Lorange. 2015. Length-weight relationships for common juvenile fishes and prey species in Hawaiian estuaries. *Journal of Applied Ichthyology*. Pp. 1-4. <https://doi.org/10.1111/jai.12957>
- Non-refereed:**
- Nguyen, K. H., **A. K. Akiona**, C. C. Chang, V. B. Chaudhary, S. J. Cheng, S. M. Johnson, S. S. Kahanamoku, A. Lee, E. E. de Leon Sanchez, L. M. Segui, and R. L. Tanner. 2021. Who are we? Highlighting Nuances in Asian American Experiences in Ecology and Evolutionary Biology. *Bulletin of the Ecological Society of America* 00(00):e01939. <https://doi.org/10.1002/bes2.1939>
- Akiona, A.** "Thinking of traveling to Hawaii during the COVID-19 pandemic? Don't." *The Sacramento Bee*, 25 Nov. 2020.
- Akiona, A.** "Thinking of traveling to Hawaii during the pandemic? Don't. You're putting native Hawaiians at risk." *The San Diego Union-Tribune*, 23 Nov. 2020.
- Akiona, A.**, K. Wilson. "Some Southern Californians Don't Get a Day at the Beach." *The San Diego Union-Tribune*, 14 May 2020, p. B9.
- Akiona, A.** "Out with a Roar: Invasive lionfish mitigation on South Caicos." *Times of the Islands Magazine*, March 2016, pp. 26-28.

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de Fontaubert, C., H. Hertler, A. Henderson, **A. Akiona**. "Keeping Tabs on Turtles: Beach profile monitoring for marine turtle nesting areas." *Times of the Islands Magazine*, July 2015, p. 26-30.

Presentations

- 2025 Indo-Pacific Fish Conference, Taipei, Taiwan (two talks)
- 2023 Panelist, Ocean Pavilion at COP28, Sharm el Sheikh, Egypt
- 2022 Reef Futures Conference, Key Largo, FL
- 2022 International Coral Reef Symposium, Bremen, Germany
- 2022 Ocean Sciences Meeting (virtual)
- 2021 Marine Biology student seminar series, Scripps Institution of Oceanography, San Diego, CA
- 2021* University of Guam STEM Conference, Mangilao, Guam (virtual)
- 2020 Women and Minorities in Science student seminar, Scripps Institution of Oceanography, San Diego, CA
- 2020 Marine Biology second year student seminar, Scripps Institution of Oceanography, San Diego, CA (virtual)
- 2020 Marine Biology student seminar series, Scripps Institution of Oceanography, San Diego, CA
- 2019 Marine Biology departmental seminar, Scripps Institution of Oceanography, San Diego, CA
- 2018 5th International Marine Conservation Congress, Kuching, Malaysia
- 2018 43rd Annual Albert L. Tester Symposium, University of Hawaii, Honolulu, HI
- 2017 Science Night at He'eia Fishpond Kaneohe, HI
- 2017 42nd Annual Albert L. Tester Symposium, University of Hawai'i at Mānoa, Honolulu, HI
- 2013 Senior Seminar, University of San Diego, San Diego, CA
- 2013 Pacific Internship Program for Exploring Science Symposium, Hilo, HI
- 2013 School for Field Studies Center for Marine Resource Studies Symposium, South Caicos, TCI
- 2012 Pacific Internship Program for Exploring Science Symposium, Hilo, HI

*Invited speaker

Teaching and guest lectures

- 2025* SIOB 272 Advanced Statistical Technique, Generalized Additive Models (GAMs) Guest Lecturer
- 2025* SIOB 286 Sustainable Seafood, Indigenous Fishing Practices Panelist
- 2024 SIOB 272 Advanced Statistical Technique, Teaching Assistant
 - Led discussions twice a week teaching statistics and coding in R to 30 graduate students
 - Created and taught a lecture on Generalized Additive Models (GAMs)
 - Administered and graded homeworks and exams
- 2024, 2023* UCSD Summer Springboard, Guest Lecturer
- 2023 One Reef Survey Methodologies Course, Fish Identification Lecturer
- 2022* SIO90 Undergraduate Seminar, Guest Lecturer

*Invited speaker

Awards and Grants

ARCS Foundation Fellowship Award (2021-2024)	\$30,000
National Science Foundation Graduate Research Fellowship (2018-2021)	\$138,000
Tribal Membership Initiative Fellowship (2018-2025)	\$101,000
Fernando Gabriel Leonida Memorial Scholarship (2017-2018)	\$1,500
Hau'oli Mau Loa Graduate Assistantship (2016-2018)	\$83,824
Kamehameha Schools 'Imi Na'auao Scholarship (2016-2018)	\$24,000

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President's Scholarship, University of San Diego (2010-2014) \$80,000
Departmental Honors in Marine Science University of San Diego (2014)
Dean's List, University of San Diego (2010-2014)

Peer Reviews

Oecologia
PLOS One
Scientific Data
Journal of Applied Ichthyology
Coral Reefs
Thalassas
Hawai'i Sea Grant

Diversity, Equity, and Inclusion

Women and Minorities in Science, Co-Director and GUIS Coordinator (La Jolla, CA) **Jan 2019 – Jan 2025**

- Served as Co-Director (2021-2024) and Growing Up in Science (GUIS) Coordinator (2020-2021) for WMIS, a student organization dedicated to fostering a more inclusive scientific community.
- Was part of the WMIS leadership team that won the UCSD Inclusive Excellence Award in 2021.

SIO-ASK, Coordinator (La Jolla, CA) **Sep 2021 – Dec 2023**

- Founding member and mentor for the SIO-ASK (Applicant Support and Knowledge-base) program, which paired prospective BIOPC Scripps applicants with a current graduate student to guide them through the application process.
- Reviewed and provided feedback on student applications to improve their chances of admission.

Diversity Advisory Committee, Member (La Jolla, CA) **Sep 2020 – Aug 2021**

- Provided recommendations and counsel to the Scripps Director and administration on ways to advance student, staff, and faculty diversity to meet University goals.
- Provided a report to Scripps leadership outlining specific areas and steps for improvement.

Diversity Admissions Committee, Member (La Jolla, CA) **Dec 2019 – Mar 2021**

- Served on a team of Scripps students, faculty, and staff to identify a diverse pool of potential students that were likely to succeed in graduate school.
- Reviewed the Scripps admissions process and provided a report to the administration with recommended changes.

Mentorship

2022. SIO-ASK. Amanda Frank (CSU San Marcos)
2021. SIO-ASK. Rianon Wallace-Demby (University of Kansas)
2021. Científico Latino Graduate Student Mentorship Initiative. Lady Volmar (Savannah State University)
2020. Científico Latino Graduate Student Mentorship Initiative. Katelynn Warner (University of Vermont)
2020. SIO Pop-Up REU. Ahmyia Cacapit (University of Guam), Daeyla' Boyd (CSU Monterey Bay)

Skills and Certifications

- Diving: AAUS Scientific Diver (rescue, advanced, nitrox), PADI Divemaster, PADI Deep specialty, current DAN

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CPR and first aid certification, Pacific and Caribbean fish species ID, photogrammetry, traditional coral reef survey methods (i.e., belt and line intercept transects, photoquadrats, etc.), BSAC Compressor Operation certification, full face mask training. Over 700 dives total.

- Boating: USFWS certified motorboat operator (MOCC), BSAC Boat Handling certification.
 - Hard skills: R, MATLAB, Viscore, JMP, CPCe, Google Earth, Microsoft Office.
 - Soft skills: Scientific writing, public speaking, science communication and outreach, teaching and mentoring.
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Volunteer Experience

Birch Aquarium, Volunteer Diver (La Jolla, CA)

Nov 2019 – Present

- Dive in the kelp tank 2-3 times a month conducting dive shows, feeding fish, and cleaning the tank.

San Diego Humane Society, Dog Foster (San Diego, CA)

Apr 2021 – May 2022

- Provided temporary care, training, and socialization for large rescue dogs to prepare them for adoption.
- Coordinated veterinary visits and adoption logistics.

Hawaiian Humane Society, Volunteer (Honolulu, HI)

Sep 2016 – Sep 2018

- Assisted at adoption events, facilitating meet-and-greets with potential adopters and providing information on pet care and animal behavior.
- Provided exercise and enrichment through regular dog walking, improving socialization and adoptability.
- Assisted in bathing and grooming dogs to maintain their health and cleanliness.